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Effects of trace metal supplement form on markers of epithelial barrier competency during repeated intramammary challenges

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ABSTRACT

Mastitis negatively affects milk composition, partly due to damage to mammary secretory epithelium. Trace metals are important cofactors for oxidative balance, immune cell function, and maintaining epithelial integrity. The objective of this study was to evaluate blood and milk indices of epithelial barrier integrity in cows supplemented with sulfate or methionine hydroxy analog chelate of Zn, Cu, and Mn at isometal levels during sterile mastitis. A total of 36 mid-lactation multiparous Holstein cows were used. Cows received either (1) sulfate-bound trace metals and intramammary infusions of saline (SULF-CON, n = 9), (2) sulfate-bound trace metals and intramammary infusions of formalin-fixed *Staphylococcus aureus* (SULF-CHAL, n = 12), or (3) methionine hydroxy analog chelates of Zn, Cu, and Mn and intramammary infusions of formalin-fixed *Staph. aureus* (MTX-CHAL, n = 12). The study lasted 6 wk, and cows were repeatedly intramammarily infused every 3 d during the final 2 wk to elicit chronic mastitis. Milk composition was assessed at each milking throughout the challenge period, and blood composition was evaluated during the final challenge. The MTX-CHAL and SULF-CHAL cows had increased milk lactate and plasma lactose content. Milk concentrations of Na, S, and Mg increased, whereas Zn, Mn, P, and Fe decreased throughout the challenge period in MTX-CHAL and SULF-CHAL cows. Plasma concentrations of Zn, P, Fe, and Mg decreased during the final challenge in MTX-CHAL and SULF-CHAL cows. Results indicate that repeated infusions of formalin-fixed *Staph. aureus* did compromise epithelial barrier integrity; however, trace metal supplement form had minimal effect on indices of epithelial function.

Key words: zinc, iron, milk minerals, plasma minerals

INTRODUCTION

Mastitis significantly alters milk composition. Concentrations of lactose and milk-derived proteins are typically reduced (Schultz, 1977; Stelwagen et al., 1995), whereas concentrations of electrolytes and serum-derived proteins are increased (Bruckmaier et al., 2004; Wellnitz et al., 2015) in milk from affected quarters. Changes in milk composition coincide with neutrophil diapedesis from blood into milk during the early inflammatory response (Harmon, 1994). Disruption of the mammary epithelial barrier is commonly attributed to the prolonged opening of tight junctions driven by sustained neutrophil infiltration (Kobayashi et al., 2013); however, epithelial cell loss and exfoliation may also compromise barrier competency (Hannon and Heald, 1982). When present in large numbers, neutrophils can exacerbate barrier damage through oxidative damage associated with reactive oxygen species. Although reactive oxygen species are essential for pathogen clearance, excessive production can damage mammary epithelial cells (Sordillo and Aitken, 2009). Compromise of the epithelial barrier allows unregulated passage of blood constituents into milk and the concurrent loss of milk components (Wellnitz and Bruckmaier, 2021).

Trace metals such as zinc (Zn), copper (Cu), and manganese (Mn) are essential for supporting immune-related functions and maintaining oxidative balance (Miller et al., 1993; Spears and Weiss, 2008), both of which can indirectly influence the barrier integrity of epithelial structures. The bioavailability of trace metals varies by source and form fed. Trace metals are commonly supplemented as inorganic salts, but their bioavailability can be low (NASEM, 2021). Organic-complexed trace metals may offer a more bioavailable alternative to inorganic sources, as they can increase intestinal absorption (Predieri et al., 2005; Wright et al., 2008). We previously reported that cows being fed methionine hydroxy analog chelates of Zn, Cu, and Mn had a greater total antioxidant status than cows that were fed sulfate-bound trace metals after repeated challenges with formalin-fixed *Staphylococcus*

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aureus (Montgomery et al., 2025). Additionally, cows fed methionine hydroxy analog chelates of Zn, Cu, and Mn had greater milk lactose concentrations than cows fed sulfate-bound trace metals after mastitis challenge, suggesting there was less leakage of milk lactose into blood, which may be indicative of better epithelial barrier competency. It remains unknown, however, what degree of lactose leakage occurred and if metal ions and complexes from blood and interstitial fluid were transversing the mammary epithelium into milk, and vice versa. Additionally, if changes in milk mineral content were occurring, determining if changes were a result of leakage or changes in blood metal concentrations was also unknown.

Our first study objective was to quantify the effect of repeated formalin-fixed *Staphylococcus aureus* challenges on mammary epithelial barrier competency by assessing markers of leakage in blood and milk. Our second objective was to assess the influence of form of trace metal supplementation on these blood and milk markers to determine if the form of the trace metals fed alters leakage of milk components into blood and vice versa. The hypothesis tested was that cows challenged with formalin-fixed *Staph. aureus* would experience compromised barrier function, and that cows supplemented with methionine hydroxy analog chelates of Zn, Cu, and Mn would experience less leakage during chronic mastitis compared with those receiving sulfate-bound Zn, Cu, and Mn.

MATERIALS AND METHODS

Animal Selection and Study Design

Details of the animals used and the experimental design have been described in full previously (Montgomery et al., 2025). Briefly, 36 multiparous Holstein cows between 60 and 140 DIM were used. Cows were clinically healthy and had an SCC less than 200,000 cells/mL. Cows were housed in individual tiestalls and allowed a 7-d acclimation period before the study commenced. Cows were blocked by DIM and milk yield into 12 blocks, where the 3 cows in each block were randomly assigned to 1 of 3 treatments. Cows were then subject to the 6-wk experimental timeline depicted in Figure 1A. Energy and trace metal availability requirements were calculated using the NASEM (2021) recommendations for maintenance and production based on the average herd BW, milk yield, and DMI of mid-lactation cows. The diet containing sulfate-bound Zn, Cu, and Mn was formulated to provide 339.5, 12.9, and 3.4 mg of available Zn, Cu, and Mn per day, respectively. The diet containing methionine hydroxy analog chelates of Zn, Cu, and Mn (Mintex, Novus International, Inc., Chesterfield) with repeated intramammary infusions of formalin-fixed *Staph. aureus* (**MTX-CHAL**) was correspondingly formulated to be iso-metal and iso-methionine relative to the sulfate-bound trace metal diet. During the first week of the study, all cows consumed the same basal sulfate diet, which was top-dressed with trace metal antagonists to reduce trace metal absorption (detailed in Montgomery et al., 2025). Thereafter, on d 0, cows began consuming their respective treatment diets for the study's duration. During the final 2 wk of the study, cows were repeatedly administered intramammary infusions of either 20 mL of sterile phosphate-buffered sterile saline (cat. # SH30256.01, Cytiva, Marlborough, MA) or 2 billion cfu of formalin-fixed *Staphylococcus aureus* suspended in 20 mL of the same saline in all the rear quarters on d 28, 31, 34, 37, and 40. All cows that received the repeated infusions of formalin-fixed *Staphylococcus aureus* developed moderate clinical mastitis after the second or subsequent challenges (i.e., visually abnormal milk and infrequent swelling and redness of mammary quarters); the resulting SCS from Montgomery et al. (2025) are represented in Figure 1B. The 3 unique ration-intramammary infusion treatment combinations were (1) control cows fed a diet containing sulfate-bound trace metals that were repeatedly intramammarily infused with saline (**SULF-CON**, n = 12); (2) cows fed a sulfate-bound trace metal diet that were repeatedly intramammary challenged with formalin-fixed *Staph. aureus* (**SULF-CHAL**, n = 12); and (3) cows fed an MTX-CHAL diet (n = 12) containing methionine hydroxy analog chelates of Zn, Cu, and Mn (Mintex, Novus International, Inc., Chesterfield) that received repeated intramammary infusions of formalin-fixed *Staph. aureus*. Jugular blood samples were collected from all cows the day preceding the initial challenge (d 27) and again on d 40 and 41 at 0, 3, 6, 9, 12, 18, 24, 30, and 36 h postfinal intramammary infusion. Cows were milked 2×/d and fed 1×/d, having ad libitum access to water and feed throughout the study. Milk samples were collected weekly (d 0 to 27) and daily (d 28 to 41) at both milkings. Three cows from the SULF-CON group developed mastitis, likely a result of saline propelling resident bacteria/debris in streak canal into the mammary gland; all data associated with these cows were excluded from the study.

Milk Collection and Analysis

Whole milk was collected and processed immediately postmilking. A portion of each sample was used to isolate the skim fraction by centrifugation at 2,000 × *g* for 20 min at 4°C, and the resulting skim milk was aliquoted into 1.5 mL microcentrifuge tubes for determination of milk L-lactate and milk glucose content. A separate 5 mL whole milk aliquot was obtained for determination

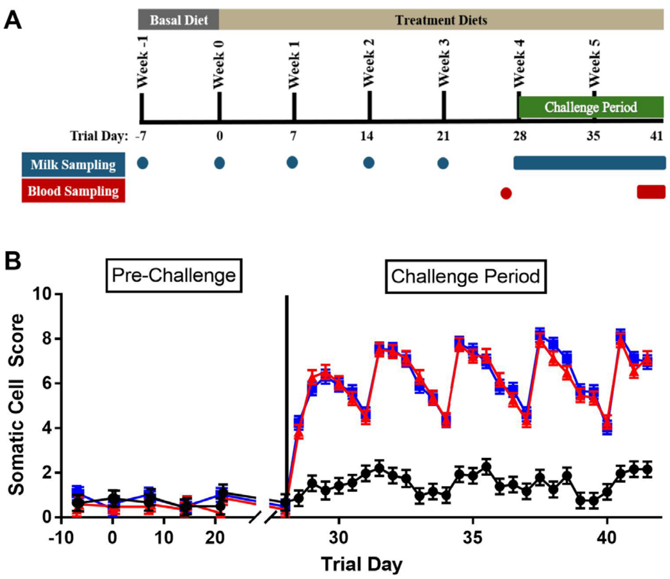

Figure 1. Experimental timeline for the 36 cows used in this study (A). The timeline illustrates the periods during which the basal and treatment diets were administered, the start of the intramammary challenge period, and the collection points for milk and blood samples. Each black line represents the trial week and corresponding trial day. Mean SCS represented from Montgomery et al. (2025; B) of cows receiving either sulfate-bound trace metals and repeated intramammary infusions of saline (SULF-CON, black circles, n = 9), sulfate bound trace metals and repeated intramammary infusions of formalin-fixed *Staphylococcus aureus* (SULF-CHAL, red triangles, n = 12), or chelated trace metals and repeated intramammary infusions of formalin-fixed *Staph. aureus* (MTX-CHAL, blue squares, n = 12) before the challenge period (d −7 to d 27) and during the challenge period (d 28 to d 41). Error bars denote SEM.

of milk metal content. All samples were stored at −20°C until analyzed.

L-Lactate was quantified in skim milk using a fluorometric assay described by Gammariello et al. (2024), who used adapted procedures from Shangraw et al. (2020). Briefly, solution A consisted of β-NAD^+ (cat. # N0632, Sigma, Ronkonkoma, NY) and L-lactate dehydrogenase from bovine heart (cat. # L3916, Sigma) in phosphate buffer. Solution A was combined with undiluted skim milk samples in a black 96-well plate in duplicate and incubated at 37°C for 30 min. Solution B consisted of resazurin (cat. # 199303, Sigma) and diaphorase enzyme from *Clostridium kluyveri* (Sigma; cat. # D2197) in a buffer solution. Solution B was then added to the plate and incubated at room temperature for 30 min. Fluorescence was measured at an excitation/emission wavelength of 530/590 nm. Intra-assay and interassay covariance of variations were 5.3% and 3.3%.

Glucose was measured in skim milk using a fluorometric assay adapted from Larsen (2015). Solution A consisted of hexokinase enzyme (1,500 IU/mL; Roche, Indianapolis, IN) and glucose-6-phosphate dehydrogenase (1,000 IU/mL; Roche; cat. # 10165875001) in a buffer solution. Solution A was added to undiluted skim milk samples in a black 96-well plate in duplicate and incubated at 37°C for 30 min. Solution B consisted of resazurin (cat. # 199303, Sigma) and diaphorase enzyme from *Clostridium kluyveri* (Sigma; cat. # D2197) in a buffer solution. Solution B was then added to the plate and incubated at room temperature for 30 m. Fluorescence was measured at an excitation/emission wavelength of 544/590 nm. Intra-assay and interassay covariance of variations were 3.0% and 2.7%.

Whole milk (2 mL) was digested in 5 mL of nitric acid and 0.5 mL of hydrochloric acid. Metal concentrations of the digested whole milk were then determined by inductively coupled plasma spectroscopy by the Service Testing and Research (**STAR**) Laboratory (The Ohio State University, Wooster, OH).

Blood Collection and Analysis

Collected jugular blood was immediately placed on ice. Blood samples were centrifuged at 1,600 × *g* for 15 min at 4°C to isolate the plasma fraction, which was aliquoted into 5 mL tubes for trace metal quantification and 1.5 mL microcentrifuge tubes for plasma lactose analysis. All aliquots were stored at −20°C until analysis.

Plasma lactose was quantified using a spectrophotometric assay (K-LOLAC assay kit, cat. # 700004314, Megazyme, Lansing, MI) according to the manufacturer's instructions. Briefly, reagent A consisted of glucose oxidase and catalase in a buffer solution. Reagent A and hydrogen peroxide were added to the samples to enzymatically remove glucose content. Samples were boiled for 7 min in 15 mL conical tubes and centrifuged at 4,000 × *g* for 25 min at 21°C. The resulting supernatant was transferred to a clear 96-well plate in duplicate and combined with reagent B, which consisted of NADP+ and ATP in a buffer solution. Absorbance was measured at 340 nm, and D-lactose concentrations were calculated based on the stoichiometric formation of 2 moles of NADPH per mole of D-glucose released during lactose hydrolysis. Intra-assay and interassay covariance of variations were 3.7% and 8.7%.

Plasma (2 mL) was digested in 5 mL of nitric acid and 0.5 mL of hydrochloric acid. Metal content of the digested plasma were then determined by inductively coupled plasma spectroscopy at the STAR Laboratory (The Ohio State University, Wooster, OH).

Statistical Analysis

All response variables were analyzed using PROC GLIMMIX in SAS 9.4 (SAS Institute). The fixed explanatory effects were treatment, time, and their interaction; random effects were block and cow nested within block. Kenward and Roger (1997) denominator df adjustment was used for all models. When measures were repeated on cows over time, the following covariance structures were examined for each response variable: compound symmetry, heterogeneous compound symmetry, first-order autoregressive, heterogeneous first-order autoregressive, and Toeplitz. When time point intervals were uneven, only variance components, compound symmetry, and heterogeneous compound symmetry covariance structures were tested. Covariance structures were selected for each separate response variable based on the lowest corrected Akaike's information criterion values (Hurvich and Tsai, 1989). Least squares means resulting from the models were contrasted using Fisher's Least Significant Difference test. Differences were considered significantly different if $P \leq 0.05$ and a tendency if $0.05 < P \leq 0.15$.

RESULTS

Milk Alterations

Glucose. Milk glucose concentrations differed between treatments at the milking before the first intramammary challenge ($P = 0.05$) and were therefore used as a covariate to assess treatment effects in subsequent analysis. Milk glucose concentration was similar among treatments following the first intramammary infusion ($P = 0.18$). Mean milk glucose concentrations were 0.34 ± 0.02 m*M* for MTX-CHAL, 0.38 ± 0.02 m*M* for SULF-CHAL, and 0.40 ± 0.03 m*M* for SULF-CON cows during the first challenge period. Similarly, milk glucose concentrations did not differ among treatments after the final challenge ($P = 0.38$); mean milk glucose concentrations were 0.35 ± 0.02 m*M* for MTX-CHAL, 0.38 ± 0.02 m*M* for SULF-CHAL, and 0.38 ± 0.03 m*M* for SULF-CON cows.

Lactate. Milk lactate content was similar among treatments immediately preceding the first challenge (i.e., h = 0; $P = 0.58$; Figure 2). Following the first challenge, milk lactate concentrations increased in MTX-CHAL ($P < 0.01$; 133 ± 12.4 μ*M*) and SULF-CHAL ($P = 0.02$; 121 ± 12.4 μ*M*) cows and were greater than SULF-CON cows (80 ± 14.1 μ*M*), with no differences between MTX-CHAL and SULF-CHAL cows ($P = 0.45$). Milk lactate concentrations spiked earlier and to a markedly greater degree in MTX-CHAL and SULF-CHAL cows during the final challenge than during the initial challenge; both having greater milk lactate concentrations than SULF-CON cows ($P < 0.01$); MTX-CHAL and SULF-CHAL cows did not differ in milk lactate content ($P = 0.47$).

Zinc. Milk Zn concentrations were greater in MTX-CHAL and SULF-CHAL cows than SULF-CON cows immediately preceding the first challenge period ($P = 0.02$) and therefore used as a covariate to assess treatment effects in subsequent analysis. Milk Zn concentrations (Figure 3A) were affected by treatment ($P \leq 0.01$) and marginally affected by the interaction between treatment and time in response to the first challenge ($P = 0.12$). Specifically, Zn concentrations in MTX-CHAL and SULF-CHAL cows decreased to concentrations below those of SULF-CON cows at 24 h postchallenge ($P \leq 0.01$) and returned to concentrations comparable to SULF-CON cows by 60 h after the initial challenge ($P = 0.57$). Milk Zn concentrations did not differ between MTX-CHAL and SULF-CHAL during the initial challenge ($P = 0.56$). Following the final intramammary challenge (Figure 4A), milk Zn concentrations were marginally affected by treatment ($P = 0.13$) and were affected by the interaction between treatment and time ($P \leq 0.01$). Zinc concentrations decreased in MTX-CHAL and SULF-CHAL cows 24 h post the final infusion to concentrations below SULF-CON cows ($P \leq 0.01$) and regained normal concentrations by h 36 ($P = 0.47$). Milk Zn concentrations did not differ between MTX-CHAL and SULF-CHAL during the final challenge ($P = 0.38$).

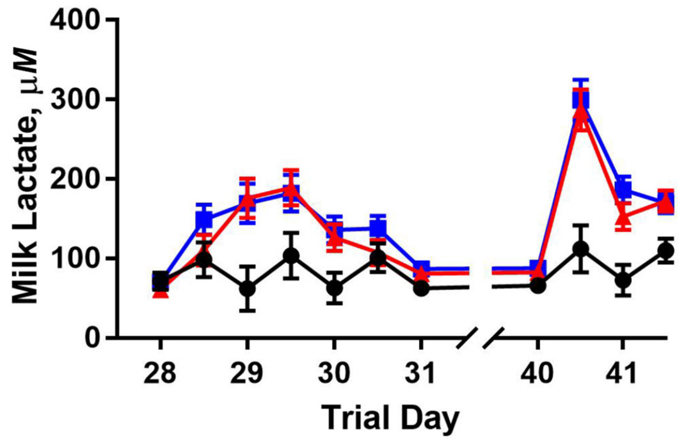

Figure 2. Milk lactate concentrations of the first (d 28–31) and final (d 40–41) intramammary challenge of cows that received either sulfate-bound trace metals and repeated intramammary infusions of saline (SULF-CON, black circles, n = 9), sulfate-bound trace metals and repeated intramammary infusions of formalin-fixed *Staphylococcus aureus* (SULF-CHAL, red triangles, n = 12), or chelated trace metals and repeated intramammary infusions of formalin-fixed *Staph. aureus* (MTX-CHAL, blue squares, n = 12). Error bars denote the SEM.

Copper. Milk Cu concentrations were marginally affected by treatment interacting with time during the initial intramammary challenge ($P = 0.11$; Figure 3B). Milk Cu concentrations did not differ among treatment groups immediately preceding challenge ($P = 0.79$), but milk Cu concentrations of MTX-CHAL cows were intermittently greater than that of SULF-CHAL cows but not SULF-CON cows (Figure 4B). In response to the final challenge, milk Cu content responses were variable and did

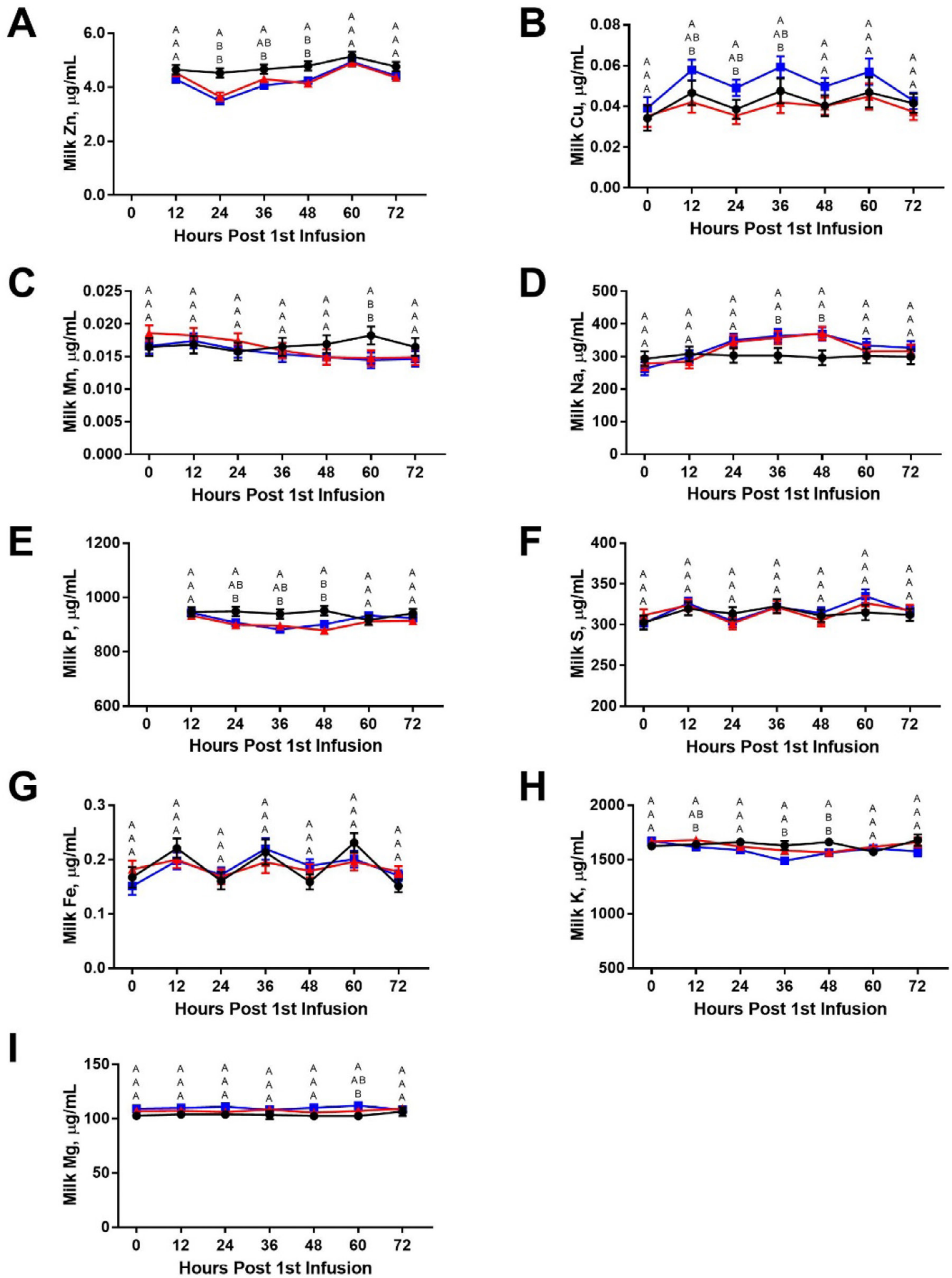

Figure 3. Whole milk mineral concentrations during the first intramammary challenge period (d 28–31) of cows that received either sulfate-bound trace metals and intramammary infusions of saline (SULF-CON, black circles, n = 9), sulfate-bound trace metals and intramammary infusions of formalin-fixed *Staphylococcus aureus* (SULF-CHAL, red triangles, n = 12), or chelated trace metals and intramammary infusions of formalin-fixed *Staph. aureus* (MTX-CHAL, blue squares, n = 12) during the first intramammary challenge. Baseline prechallenge concentrations of Zn and P were used as a covariate. Error bars denote the SEM. Error bars are not observable for some means because the SEM is smaller than the symbol's size.

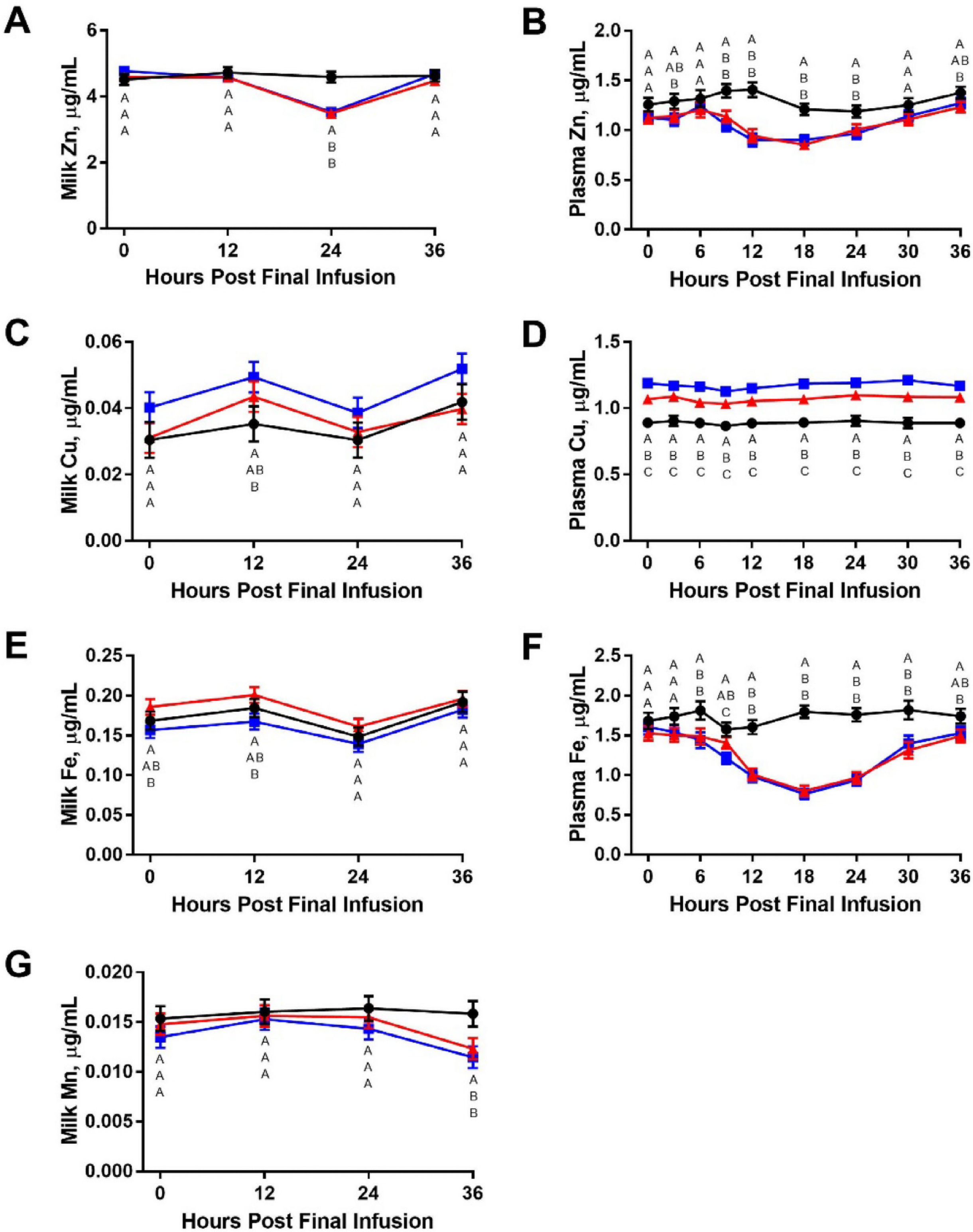

Figure 4. Whole milk and plasma micromineral concentrations of cows that received either sulfate-bound trace metals and intramammary infusions of saline (SULF-CON, black circles, n = 9), sulfate-bound trace metals and intramammary infusions of formalin-fixed *Staphylococcus aureus* (SULF-CHAL, red triangles, n = 12), or chelated trace metals and intramammary infusions of formalin-fixed *Staph. aureus* (MTX-CHAL, blue squares, n = 12) during the final intramammary challenge. Baseline prechallenge concentrations of Zn were used as a covariate. Error bars denote the SEM. Error bars are not depicted for some means because the SEM is smaller than the symbol's size.

not differ among treatments except 12 h postchallenge where MTX-CHAL cows exhibited a greater Cu concentration than SULF-CON cows (Figure 4C; $P = 0.05$).

Manganese. Milk Mn content was comparable among treatments before the challenge period ($P = 0.34$) but was affected by the interaction between treatment and time ($P < 0.01$; Figure 3C). Specifically, Mn concentrations in milk from MTX-CHAL and SULF-CHAL cows declined by 60 h postinitial challenge and were lower than SULF-CON cows ($P < 0.05$). Milk Mn concentrations responded similarly during the final challenge, being affected by treatment interacting with time ($P < 0.01$; Figure 4G). Milk Mn content was similar among treatment groups at final infusion but became reduced in MTX-CHAL and SULF-CHAL cows by 36 h postchallenge relative to SULF-CON cows ($P \leq 0.01$).

Sodium. Milk Na content was marginally affected by treatment interacting with time during the initial challenge ($P = 0.08$; Figure 3D); Na content was similar among treatment groups immediately before intramammary infusion at h 0 ($P = 0.47$) but was greater in MTX-CHAL (329 ± 14.7 μg/mL) and SULF-CHAL cows (324 ± 14.6 μg/mL) than SULF-CON cows (300 ± 15.6 μg/mL) at 36 and 48 h postchallenge ($P \leq 0.03$). Milk Na similarly increased during the final challenge in response to formalin-fixed *Staph. aureus* administration (Figure 5C); milk Na concentrations of MTX-CHAL cows (320 ± 11.9 μg/mL) and SULF-CHAL cows (308 ± 11.8 μg/mL) were greater than that of SULF-CON cows (275 ± 13.8 μg/mL; $P = 0.01$ and $P = 0.07$, respectively). Mean milk Na content of MTX-CHAL and SULF-CHAL cows did not differ during the final challenge ($P = 0.45$).

Phosphorus. Milk P content was greater in SULF-CHAL ($P = 0.05$) and marginally greater in MTX-CHAL ($P = 0.09$) compared with SULF-CON cows immediately preceding the first challenge period and therefore used as a covariate to assess treatment effects in subsequent analysis. During the first challenge (Figure 3E), milk P content was not affected by treatment ($P = 0.16$) but was affected by the interaction between treatment and time ($P \leq 0.01$). Milk P concentrations began to decline in MTX-CHAL (915 ± 11.5 μg/mL) and SULF-CHAL cows (905 ± 11.6 μg/mL), falling below levels observed in SULF-CON cows (941 ± 13.6 μg/mL) at 24 h after the initial challenge ($P \leq 0.04$). However, there were no differences in milk P content following the final intramammary challenge ($P = 0.80$; Figure 5E).

Sulfur. Milk S content did not differ among treatments at the first challenge ($P = 0.47$). After the first intramammary challenge (Figure 3F), milk S was not affected by treatment alone ($P = 0.89$) but tended to increase over time in MTX-CHAL and SULF-CHAL cows, as indicated by a treatment-by-time interaction ($P = 0.08$). Similarly, following the final challenge (Figure 5G), overall milk S content was not different among treatments ($P = 0.23$), but the treatment-by-time interaction was significant ($P < 0.01$), with MTX-CHAL and SULF-CHAL cows showing modest increases in milk S.

Iron. Milk Fe content did not differ between treatments before the challenge period ($P = 0.41$) or during the first challenge ($P = 0.99$; Figure 3G). During the final challenge (Figure 4E), milk Fe content of MTX-CHAL ($P = 0.25$; 0.16 ± 0.007 μg/mL) and SULF-CHAL cows ($P = 0.23$; 0.19 ± 0.007 μg/mL) did not differ from SULF-CON cows (0.17 ± 0.008 μg/mL), but Fe content of SULF-CHAL cows was greater than that of MTX-CHAL cows ($P = 0.01$).

Potassium. Milk K concentrations were similar among treatments before the first challenge ($P = 0.48$). During the first challenge (Figure 3H), milk K content was significantly affected by a treatment-by-time interaction ($P < 0.01$); K concentrations were marginally lower in MTX-CHAL and SULF-CHAL cows after challenging than in SULF-CON cows ($P = 0.12$). During the final challenge (Figure 4E), milk K content was not affected by treatment ($P = 0.73$) or treatment interacting with time ($P = 0.95$).

Magnesium. Milk Mg content did not differ between treatments immediately before the challenge ($P = 0.32$) or during the first intramammary challenge ($P = 0.32$; Figure 3I). During the final challenge, however, milk Mg content was greater in MTX-CHAL cows ($P = 0.01$; 111 ± 2.7 μg/mL) and was marginally greater in SULF-CHAL cows ($P = 0.09$; 108 ± 2.7 μg/mL) compared with SULF-CON cows (102 ± 3.1 μg/mL), with no difference between MTX-CHAL and SULF-CHAL cows ($P = 0.29$; Figure 5A).

Blood Alterations

Plasma metal content was similar among treatments the day preceding the start of the challenge period (Table 1). Responses of plasma metal content during the final challenge are depicted in Figures 4 and 5 if a marginal or significant treatment response or treatment-by-time response was detected. Plasma Na, S, and K content did not differ between treatments during the final challenge ($P > 0.15$) and are not described in text. Plasma Mn content was below the limit of detection and therefore is not included.

Zinc. Plasma Zn concentrations during the final intramammary challenge are presented in Figure 4B. Plasma Zn content was lower in MTX-CHAL ($P < 0.01$; 1.08 ± 0.04 μg/mL) and SULF-CHAL cows ($P < 0.01$; 1.10 ± 0.04 μg/mL) compared with SULF-CON cows, with no differences between MTX-CHAL and SULF-CHAL cows ($P = 0.78$).

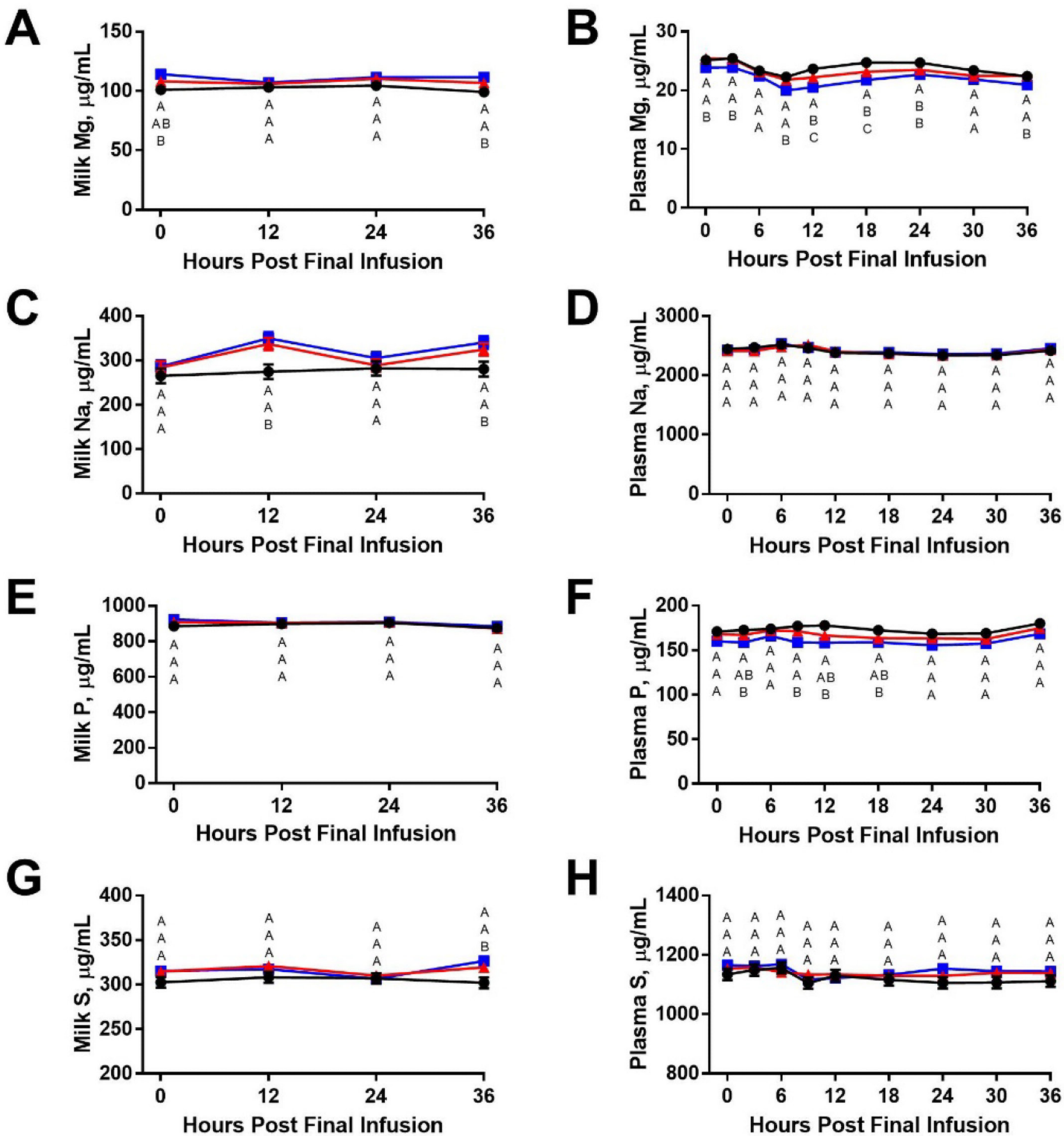

Figure 5. Whole milk and plasma macromineral concentrations of cows that received either sulfate-bound trace metals and intramammary infusions of saline (SULF-CON, black circles, n = 9), sulfate-bound trace metals and intramammary infusions of formalin-fixed *Staphylococcus aureus* (SULF-CHAL, red triangles, n = 12), or chelated trace metals and intramammary infusions of formalin-fixed *Staph. aureus* (MTX-CHAL, blue squares, n = 12) during the final intramammary challenge. Baseline prechallenge concentrations of P were used as a covariate. Error bars denote the SEM. Error bars are not depicted for some means because the SEM is smaller than the symbol's size.

Copper. Plasma Cu concentrations during the final challenge are shown in Figure 4D. Plasma Cu content

Table 1. Prechallenge plasma metal content of cows assigned to receive trace metals bound to sulfates and repeated intramammary infusions of saline (SULF-CON, n = 9), sulfate-bound trace metals and repeated intramammary infusions of formalin-fixed *Staphylococcus aureus* (SULF-CHAL, n = 12), or chelated trace metals and intramammary infusions of formalin-fixed *Staph. aureus* (MTX-CHAL, n = 12) immediately before the challenge period

	Treatment				
Item	SULF-CON	SULF-CHAL	MTX-CHAL	SEM	*P*-value
Zn, μg/mL	1.2	1.2	1.1	0.08	0.65
Cu, μg/mL	0.85	0.87	0.91	0.03	0.38
Fe, μg/mL	1.9	1.8	1.6	0.14	0.26
Mg, μg/mL	23.7	23.7	21.7	0.77	0.40
Na, μg/mL	2,470	2,470	2,470	59.2	0.99
P, μg/mL	161	167	167	4.9	0.60
K, μg/mL	967	928	933	18.4	0.25
S, μg/mL	1,143	1,122	1,107	17.8	0.36

was highest in MTX-CHAL cows (1.15 ± 0.03 μg/mL), followed by SULF-CHAL (1.05 ± 0.03 μg/mL), then SULF-CON cows (0.16 ± 0.04 μg/mL), with all pairwise differences being significant ($P < 0.01$).

Phosphorus. Plasma P concentrations during the final challenge are shown in Figure 5F. Concentrations of plasma P were not significantly affected by treatment ($P = 0.15$) but were affected by the interaction between treatment and time ($P = 0.03$). The MTX-CHAL cows ($P = 0.06$; 160.95 ± 3.89 μg/mL) had marginally lower P concentrations than SULF-CON cows (172.34 ± 4.56 μg/mL); SULF-CHAL (167.73 ± 3.89 μg/mL) cows had similar P concentrations as MTX-CHAL ($P = 0.21$) and SULF-CON ($P = 0.43$).

Iron. Plasma Fe concentrations during the final intramammary challenge are shown in Figure 4F. Both MTX-CHAL ($P < 0.01$; 1.30 ± 0.06 μg/mL) and SULF-CHAL cows ($P < 0.01$; 1.32 ± 0.06 μg/mL) had lower plasma Fe concentrations compared with SULF-CON cows (1.74 ± 0.07 μg/mL); Fe concentrations of SULF-CHAL and SULF-CON were similar ($P = 0.71$).

Magnesium. Plasma concentrations of Mg during the final challenge are shown in Figure 5B. Plasma Mg concentrations were lower in MTX-CHAL cows ($P < 0.01$; 21.95 ± 0.37 μg/mL) compared with both SULF-CHAL (23.29 ± 0.36 μg/mL) and SULF-CON cows (23.85 ± 0.41 μg/mL), with no difference observed between SULF-CHAL and SULF-CON cows.

Lactose. Plasma lactose concentrations are shown in Figure 6. Baseline plasma lactose concentrations did not differ among treatments on the day preceding the challenge period (d 27; $P = 0.82$). Plasma lactose concentrations increased in response to intramammary infusions of formalin-fixed *Staph. aureus* and became greater in MTX-CHAL (25.19 ± 2.34 mg/L; $P < 0.01$) and SULF-CHAL (24.23 ± 2.34 mg/L; $P < 0.01$) cows than SULF-CON cows (9.96 ± 2.70 mg/L). Plasma lactose concentrations did not differ between MTX-CHAL and SULF-CHAL cows ($P = 0.74$).

DISCUSSION

Our first objective of this study was to determine whether changes in milk indicators of epithelial barrier competency are attributable to epithelial leakage or to alterations in blood composition. Our second objective was to evaluate whether the form of trace metals fed to cows influenced the degree of leakage of markers of epithelial barrier competency.

The increased milk lactate and plasma lactose content in challenged cows was expected. We and others have previously reported that milk lactate (Davis et al., 2004; Lehmann et al., 2013; Gammariello et al., 2024) and

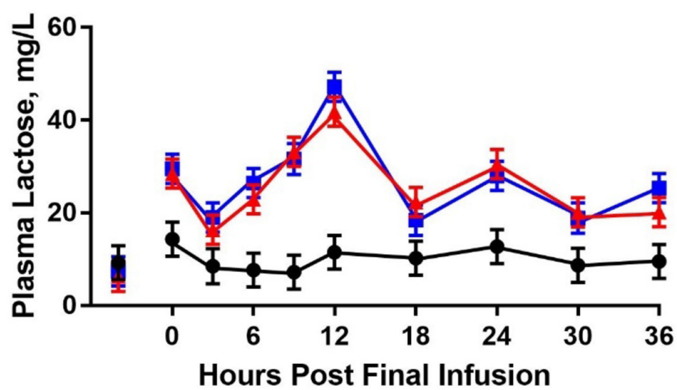

Figure 6. Plasma lactose concentrations of cows that received either sulfate-bound trace metals and intramammary infusions of saline (SULF-CON, black circles, n = 9), sulfate-bound trace metals and intramammary infusions of formalin-fixed *Staphylococcus aureus* (SULF-CHAL, red triangles, n = 12), or chelated trace metals and intramammary infusions of formalin-fixed *Staph. aureus* (MTX-CHAL, blue squares, n = 12). Mean plasma lactose concentrations are for the final intramammary challenge; the values presented before h 0 are prechallenge baseline values. Error bars denote the SEM.

plasma lactose concentrations (Johnzon et al., 2018) increase in response to mastitis challenges. Lactate is typically present at low concentrations in milk but increases during mastitis (Davis et al., 2004). Although some of this increase may result from leakage of lactate from blood into milk (Wellnitz and Bruckmaier, 2021), milk lactate concentrations can exceed those in blood (Lehmann et al., 2013), suggesting that local production by mammary epithelial cells (Silanikove et al., 2011) or leukocytes (Enger, 2019), driven by a shift to anaerobic metabolism, may also contribute to increased milk lactate concentrations. Conversely, lactose is synthesized exclusively in the mammary gland, and under healthy conditions, lactose does not cross the epithelial barrier (Linzell and Peaker, 1971). During mastitis, tight junction disruption allows lactose to leak from milk into the blood, as reflected by elevated plasma lactose in challenged cows in this study. In our previous study we observed reduced milk lactose concentrations in challenged cows (Montgomery et al., 2025), consistent with both impaired lactose synthesis and increased paracellular loss. Because glucose is the primary substrate for lactose synthesis, alterations in glucose metabolism and transport during mastitis can directly impact lactose availability in milk (Kitchen, 1981). Despite this, no differences in milk glucose concentrations were detected between challenged and control cows. This was unexpected, as neutrophils depend heavily on glucose to fuel phagocytosis and reactive oxygen species generation (Kramer et al., 2014). Activated neutrophils can uptake glucose from their environment (Borregaard and Herlin, 1982) and thus could be expected to use glucose from milk during mastitis (Enger, 2019).

Alterations in milk Na, Fe, and Zn concentrations are well-established indicators of mastitis, and changes in these metals' concentrations were expected in response to intramammary challenge (Wellnitz and Bruckmaier, 2021). In healthy mammary glands, low milk Na is maintained by intact tight junctions and active ion transport (Linzell and Peaker, 1971; Fernando et al., 1985; Schultz, 2016). During mastitis, however, tight junctions become compromised, and the loss of epithelial cells from the basement membrane reduces epithelial barrier competency, which allows the passive Na influx from the interstitial fluid into milk, resulting in elevated milk Na concentrations (Schultz, 1977). Given that plasma Na remained unchanged, the observed increase in milk Na in challenged cows can be attributed to a local disruption of epithelial barrier integrity. In contrast, Fe and Zn concentrations declined in both milk and plasma following challenge. Such reductions in Fe and Zn have been previously reported in mastitis-challenged cows (Erskine and Bartlett, 1993; Middleton et al., 2004), and the concurrent decreases in plasma levels suggest that reductions in milk Fe and Zn are not attributable to epithelial leakage but rather to redistribution associated with the host response. Zinc is essential for immune cell function and antioxidant defense (Spears and Weiss, 2008; Bonaventura et al., 2015), and reduced Zn likely reflects increased uptake by activated immune cells to support host defenses. Additionally, reductions in plasma Zn and Fe are regarded as nonspecific host defense mechanisms. Moreover, sequestration of Fe and Zn during mastitis is thought to be a nonspecific defense mechanism aimed at restricting bacterial access to nutrients required for normal growth (Weinberg, 1984; Middleton et al., 2004). Plasma Cu concentrations were greater in MTX-CHAL cows than SULF-CON cows, with SULF-CHAL cows being intermediate. This increased plasma Cu concentration during the final challenge may be a result of increased circulating ceruloplasmin, which has been reported to increase in blood during inflammatory events (Broadley and Hoover, 1989). Milk Mg increased in both challenged groups, whereas plasma Mg decreased only in MTX-CHAL cows, potentially indicating compromised epithelial integrity and increased Mg leakage from blood into milk. Concentrations of milk S tended to increase in challenged cows over time following each challenge. In previous work using a split-udder model, we observed significant increases in milk S in response to intramammary challenge (Enger and Hastings, unpublished data); however, greater between-cow variability in the present study may have limited the detection of more pronounced changes in S content. Both milk and plasma P concentrations declined following intramammary challenge. As total P was measured in this study, it is unclear whether the observed changes reflect alterations in inorganic P content, phosphates associated with casein micelles, or organic P fractions such as phospholipids in the milk fat globule membrane. Finally, milk Mn concentrations declined during the initial challenge period in challenged cows, but despite statistical significance, the magnitude of change was small and may have limited biological relevance.

We chose to evaluate both the initial and final intramammary challenges, as our previous study showed that repeated intramammary challenges led to greater immune cell recruitment and, thus, greater responses in markers of epithelial barrier competency. It is important to appreciate that this study assessed total metal content. Milk metal analysis was conducted using whole milk, which does not allow for determination of the specific distribution of minerals (e.g., magnesium bound to casein vs. free magnesium). Similarly, plasma trace metal concentrations reflect only the plasma fraction of blood; thus, trace metals contained within red blood cells and immune cells were not evaluated.

Another consideration as to how these results should be interpreted is (1) the formalin-fixed *Staph. aureus* challenge failed to elicit a detectable milk yield reduction relative to the saline-treated cows (reported in Montgomery et al., 2025) despite eliciting clinical mastitis and changes in milk composition, and (2) the study evaluated changes to epithelial barrier integrity resulting from sterile mastitis and not a live pathogen. Based on the SCS observed after each challenge, it is clear the inflammatory response began to resolve relatively soon after each challenge (within 24–36 h), and a sustained proinflammatory response was not fully maintained between each challenge. Increasing the frequency of the formalin-fixed *Staphylococcus aureus* challenges may have been needed to better sustain a proinflammatory disease state, yielding a greater disease state. Alternatively, using a live pathogen would have sustained a proinflammatory response as the bacteria grew and established an intramammary infection, and immune cells would infiltrate the mammary gland at a sustained or increasing level until the infection was cleared. Importantly though, using a live pathogen can yield highly variable responses in infection duration and severity, which was decided against because it would have complicated the goals of this study. It is important to clarify that the results of this study do not provide information on how feeding different forms of trace metals would affect mammary epithelial barrier integrity across all of the numerous mastitis pathogens, each possessing unique virulence factors that could differentially affect epithelial barrier competency.

CONCLUSIONS

The results of this study indicate that repeated intramammary infusions of formalin-fixed *Staph. aureus* compromised mammary epithelial barrier function, resulting in leakage of plasma components into milk and vice versa. Challenged cows exhibited elevated milk lactate, plasma lactose, and milk Na concentrations, along with decreased concentrations of Fe and Zn in both milk and plasma. The increased concentrations of plasma lactose and milk Na are consistent with compromised epithelial barrier function that may be resulting from a combination of epithelial cell loss and compromised tight junction integrity. The increase in milk lactate may partly reflect leakage from blood; however, local lactate production within the mammary gland, possibly due to a shift toward anaerobic metabolism by mammary epithelial cells and leukocytes, is also possible. The observed reductions in Fe and Zn levels in milk and plasma may reflect immune-mediated shifts in trace metal availability including sequestration and immune cell utilization to enhance host defense. Trace metal supplement form had little effect on these markers of epithelial barrier competency.

NOTES

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Nonstandard abbreviations used: MTX-CHAL = methionine hydroxy analog chelates of Zn, Cu, and Mn repeated intramammary infusions of formalin-fixed *Staph. aureus*; STAR = Service Testing and Research; SULF-CHAL = cows fed sulfate-bound trace metals that received intramammary infusions of formalin-fixed *Staph. aureus*; SULF-CON = cows fed sulfate-bound trace metals that received saline intramammary infusions.

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